

Week 7 Discussion: Midterm Practice (50 min)

Problems: Textbook 1.12, 2.7, 3.3, 4.18, 5.23.

Practice 1: (Textbook 1.12) Short put profit condition

A trader writes a December put option with a strike price of \$30. The price (premium) of the option is \$4. Under what circumstances does the trader make a gain?

Practice 2: (Textbook 2.7) Margin call thresholds

A trader buys two July futures contracts on frozen orange juice concentrate. Each contract is for delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is \$6,000 per contract, and the maintenance margin is \$4,500 per contract.

- (a) What **price change** would lead to a margin call?

- (b) Under what circumstances could \$2,000 be **withdrawn** from the margin account?

Practice 3: (Textbook 3.3) Optimal hedge ratio

Suppose that the standard deviation of quarterly changes in the **spot** price of a commodity is \$0.65, the standard deviation of quarterly changes in the **futures** price on the commodity is \$0.81, and the correlation between the two changes is 0.8.

- (a) What is the **optimal hedge ratio** for a 3-month contract?

- (b) What does it **mean** (interpretation in words / how you use it to pick # of contracts)?

Practice 4: (Textbook 4.18) Bond pricing and duration (continuous compounding)

A 5-year bond with a yield of 7% (continuously compounded) pays an 8% coupon at the end of each year. Assume face value is \$100 (so coupon is \$8 annually).

- (a) What is the bond's **price**?

- (b) What is the bond's **duration** (Macaulay duration, in years)?

- (c) Use duration to estimate the effect on the bond's price of a 0.2% **decrease** in yield.

- (d) Recalculate the bond's price using a 6.8% continuously compounded yield and verify agreement with (c).

Practice 5: (Textbook 5.23) Index futures with dividend yield (continuous compounding)

An index is 1,200. The 3-month risk-free rate is 3% per annum and the dividend yield over the next 3 months is 1.2% per annum. The 6-month risk-free rate is 3.5% per annum and the dividend yield over the next 6 months is 1% per annum. All interest rates and dividend yields are continuously compounded. Estimate the futures price of the index for 3-month and 6-month contracts.